and a growing cell doesn't expand its surface as fast as its volume, which mandates the maximum possible size for a single cell.

In our analogy, Harry's Hamburger Heaven can expand a single store for a while, but the size of a single store will be limited by the number of potential customers in its trading area.

*Fission.* A more efficient growth strategy that has served bacteria well to this day, fission divides the cell into two identical but separate daughter cells. The advantage of fission is conservatism, for the two cells have the same size and structure as their predecessor. The disadvantage is that the cells must remain simple in function and microscopic in size.

Our hamburger stand can model fission by opening a second store as a separate corporation and letting each run its own show.

*Chaining.* A third strategy is that of the seaweed and the sponge. Cells divide into a mass of identical daughter cells, but instead of separating, the cell walls stick together, forming a chain. This has the advantage of allowing macroscopic size, which opens up some new habitats. The disadvantage is that the chain cannot do very much more than the single cell could.

If Harry's Hamburger Heaven opens forty identical hamburger stands, it would be using a chaining strategy.

*Differentiation.* The fourth and last strategy is much more complex. Most multicellular organisms, including us, are made up of cells that have taken up different shapes and functions, allowing the development of leaves, stems, bones, fins, eyes, and all the myriad beautiful tissues and